

Ruffley & Lutz et al. — statistics computed for the NEE revision

Flowering time $\times$ WUE ($\delta^{13}\text{C}$): correlations, slopes, covariances — plus the Table S28 G-matrix constraint analysis

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Everything below is recomputed from the deposited data. Written to answer Reviewer 2 comment 1 (“no p-value or r^2 values are listed”) and the related Reviewer 1 points on Fig. 3B/3C.

The final section (*Table S28 — evolvability, respondability, and multivariate constraint*) is a walkthrough of the Hansen & Houle framework behind Table S28, including the construction of the G matrix behind it. The pipeline that generates the corresponding supplementary tables lives in `regenerated-tables/regenerate_S21_S28_S29.R`.

Source data

```
SUB <- "~/projects/natvar/NatEcoEvo_SubmissionNovember"
DAT <- file.path(SUB, "Athaliana_Natural_Selection_Conflict/data")
CHAM <- "~/projects/natvar/Ruffley_Lutz_et_al_joint_MS/Data_code_Ruffley_Lutz_et_al"

src <- data.frame(
  file = c("deltaC_extended_group_rep3.csv",
           "deltaC_extended_group_rep3_pval.csv",
           "lines_info_annot.csv",
           "atlas1001_phenotypes_matrix_MR.csv",
           "imputed_wt&mut_dc13.csv"),
```

```

where = c(CHAM, CHAM, CHAM, DAT, DAT),
what = c("Uli's  $\delta^{13}\text{C}$  experiment: 49 line means (29 flc-edited, 19 wild-type progenitors, Col-0), 3 reps each",
"The 19 edited lines that have a paired wild-type progenitor, with per-line p-values vs that progenitor.",
"Metadata for 112 edited lines: progenitor accession, FT16 from 1001G, wild-type flowering time and FLC qRT.",
"1001 Genomes phenotype matrix, MEASURED values (Delta_13C, FT10, FT16).",
"Imputed  $\delta^{13}\text{C}$  for 306 accessions, separate wild-type and flc-mutant predictions.")
knitr::kable(src[, c("file", "what")])

```

file	what
deltaC_extended_group_rep3.csv	Uli's $\delta^{13}\text{C}$ experiment: 49 line means (29 flc-edited, 19 wild-type progenitors, Col-0), 3 reps each. Fig. 3B source.
deltaC_extended_group_rep3_pval.csv	The 19 edited lines that have a paired wild-type progenitor, with per-line p-values vs that progenitor.
lines_info_annot.csv	Metadata for 112 edited lines: progenitor accession, FT16 from 1001G, wild-type flowering time and FLC qRT.
atlas1001_phenotypes_matrix_MR.csv	1001 Genomes phenotype matrix, MEASURED values (Delta_13C, FT10, FT16).
imputed_wt&mut_dc13.csv	Imputed $\delta^{13}\text{C}$ for 306 accessions, separate wild-type and flc-mutant predictions.

```

d <- read.csv(file.path(CHAM, "deltaC_extended_group_rep3.csv"))
pr <- read.csv(file.path(CHAM, "deltaC_extended_group_rep3_pval.csv"))
at <- read.csv(file.path(DAT, "atlas1001_phenotypes_matrix_MR.csv"))
im <- read.csv(file.path(DAT, "imputed_wt&mut_dc13.csv"))

# groups. dag_mean = days to flowering, 21 C chamber; deltaC_mean_mean = measured d13C
E <- subset(d, label == "mut") # 29 edited lines
W <- subset(d, label == "wt") # 19 wild-type progenitors
WC <- subset(d, label %in% c("wt", "col")) # + Col-0 (n = 20; this is the Fig 3B green fit)
at16 <- at[!is.na(at$FT16) & !is.na(at$Delta_13C), ]
at10 <- at[!is.na(at$FT10) & !is.na(at$Delta_13C), ]
c(edited = nrow(E), progenitors = nrow(W), progenitors_col0 = nrow(WC),
  atlas_FT16 = nrow(at16), atlas_FT10 = nrow(at10))

```

```

##          edited      progenitors progenitors_col0      atlas_FT16
##          29          19          20          248
##    atlas_FT10
##          261

```

Each of the 29 edited lines comes from a **distinct** accession (no pseudo-replication), and all 29 have $\delta^{13}\text{C}$ measured in triplicate. The 19 are the subset with a paired progenitor — *not* a measured-vs-imputed distinction.

```

firstnum <- function(v) as.numeric(sapply(v, function(x) {
  m <- regmatches(x, regexpr("[0-9]+", x)); if (length(m)) m else NA }))
c(edited_lines = nrow(E), unique_accessions = length(unique(firstnum(E$line))),
  all_measured_3reps = sum(E$deltaC_mean_count == 3),
  paired_subset_of_29 = all(pr$line %in% E$line))

```

```

##          edited_lines  unique_accessions  all_measured_3reps  paired_subset_of_29
##          29          29          29          1

```

Core statistics

```

stat <- function(x, y, label) {
  ok <- !is.na(x) & !is.na(y); x <- x[ok]; y <- y[ok]

```

```

ct <- cor.test(x, y); f <- lm(y ~ x)
data.frame(set = label, n = length(x),
           r = round(unname(ct$estimate), 3),
           p = signif(ct$p.value, 3),
           slope = round(unname(coef(f)[2]), 2),
           slope_SE = round(summary(f)$coef[2, 2], 2),
           cov = round(cov(x, y), 3),
           sd_d13C = round(sd(x), 3),
           sd_FT = round(sd(y), 2))
}

tab <- rbind(
  stat(pr$deltaC_mean_mean, pr$dag_mean, "19 edited (paired w/ progenitor)"),
  stat(E$deltaC_mean_mean, E$dag_mean, "29 edited (all)"),
  stat(W$deltaC_mean_mean, W$dag_mean, "19 progenitors"),
  stat(WC$deltaC_mean_mean, WC$dag_mean, "20 progenitors + Col-0"),
  stat(at16$Delta_13C, at16$FT16, "248 natural accessions (FT16)"),
  stat(at10$Delta_13C, at10$FT10, "261 natural accessions (FT10)")
knitr::kable(tab, row.names = FALSE)

```

set	n	r	p	slope	slope_SE	cov	sd_d13C	sd_FT
19 edited (paired w/ progenitor)	19	0.499	0.0298	6.15	2.59	1.331	0.465	5.74
29 edited (all)	29	0.376	0.0447	4.58	2.18	0.902	0.444	5.41
19 progenitors	19	-0.066	0.7890	-5.07	18.66	-0.722	0.377	29.09
20 progenitors + Col-0	20	0.144	0.5450	10.44	16.94	1.876	0.424	30.79
248 natural accessions (FT16)	248	0.381	0.0000	8.63	1.33	12.262	1.192	26.97
261 natural accessions (FT10)	261	0.431	0.0000	6.80	0.88	9.620	1.190	18.77

Flowering time is days-to-flowering at 21 °C for the chamber sets and FT10/FT16 for the natural accessions, so slopes are comparable in kind but not identical quantities.

20 progenitors + Col-0 reproduces the $\beta = 10.44 \pm 16.9$ reported in the manuscript, confirming the Fig. 3B wild-type line includes Col-0.

Formal tests

```

fz <- function(r1, r2, n1, n2) {
  z <- (atanh(r1) - atanh(r2)) / sqrt(1/(n1 - 3) + 1/(n2 - 3))
  c(z = round(z, 3), p = round(2 * (1 - pnorm(abs(z))), 4)) }

rbind(
  `edited vs natural (FT16)` = fz(cor(E$deltaC_mean_mean, E$dag_mean),
                                cor(at16$Delta_13C, at16$FT16), nrow(E), nrow(at16)),
  `edited vs natural (FT10)` = fz(cor(E$deltaC_mean_mean, E$dag_mean),
                                cor(at10$Delta_13C, at10$FT10), nrow(E), nrow(at10)),
  `edited vs progenitors+Col0` = fz(cor(E$deltaC_mean_mean, E$dag_mean),
                                   cor(WC$deltaC_mean_mean, WC$dag_mean), nrow(E), nrow(WC)))

##              z      p
## edited vs natural (FT16) -0.033 0.9736
## edited vs natural (FT10) -0.321 0.7482
## edited vs progenitors+Col0 0.802 0.4227

```

```
b <- rbind(data.frame(x = E$deltaC_mean_mean, y = E$dag_mean, g = "edited"),
           data.frame(x = WC$deltaC_mean_mean, y = WC$dag_mean, g = "progenitor"))
round(summary(lm(y ~ x * g, data = b))$coef, 4) # interaction = slope difference
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  163.1935   264.8553  0.6162   0.5409
## x            4.5799     8.5963  0.5328   0.5968
## gprogenitor  223.1604   423.5029  0.5269   0.6008
## x:gprogenitor 5.8605    13.9030  0.4215   0.6754
```

```
cc <- function(s) cov(s$deltaC_mean_mean, s$dag_mean)
obs <- cc(E) - cc(WC)
boot <- replicate(20000, cc(E[sample(nrow(E), replace = TRUE), ]) -
                        cc(WC[sample(nrow(WC), replace = TRUE), ]))
c(observed_difference = round(obs, 3),
  CI_lower = round(quantile(boot, .025), 3), CI_upper = round(quantile(boot, .975), 3))
```

```
## observed_difference      CI_lower.2.5%      CI_upper.97.5%
##                -0.974                -8.509                6.281
```

The variance test is the robust result. Editing collapses flowering-time variance ~32-fold while leaving $\delta^{13}\text{C}$ variance untouched.

```
rbind(d13C = c(progenitors = var(WC$deltaC_mean_mean), edited = var(E$deltaC_mean_mean)),
      FT   = c(var(WC$dag_mean), var(E$dag_mean)))
```

```
##      progenitors      edited
## d13C  0.1796541  0.1969672
## FT   947.7433865 29.2977866
```

```
var.test(WC$dag_mean, E$dag_mean)
```

```
##
## F test to compare two variances
##
## data: WC$dag_mean and E$dag_mean
## F = 32.349, num df = 19, denom df = 28, p-value = 6.439e-14
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  14.37375 77.98251
## sample estimates:
## ratio of variances
##          32.34863
```

Box's M tests the whole covariance matrix, so it rejects here because of the *variance*, not the covariance — do not cite it as evidence about covariance.

```
S1 <- cov(cbind(E$deltaC_mean_mean, E$dag_mean)); n1 <- nrow(E)
S2 <- cov(cbind(WC$deltaC_mean_mean, WC$dag_mean)); n2 <- nrow(WC); p <- 2; g <- 2
Sp <- ((n1-1)*S1 + (n2-1)*S2) / (n1 + n2 - g)
M <- (n1+n2-g)*log(det(Sp)) - ((n1-1)*log(det(S1)) + (n2-1)*log(det(S2)))
cst <- 1 - (sum(1/c(n1-1, n2-1)) - 1/(n1+n2-g)) * (2*p^2+3*p-1) / (6*(p+1)*(g-1))
c(chi2 = round(M*cst, 2), df = (g-1)*p*(p+1)/2,
  p = signif(pchisq(M*cst, (g-1)*p*(p+1)/2, lower.tail = FALSE), 3))
```

```
##      chi2      df      p
## 5.759e+01 3.000e+00 1.920e-12
```

Covariance scaling

Scaling each set by its **own** SDs returns the correlation and discards the magnitude information. Scaling all sets by a **common** reference (the natural population) keeps it.

```
sdX0 <- sd(at16$Delta_13C); sdy0 <- sd(at16$FT16)
sc <- function(x, y, lab) data.frame(set = lab, r = round(cor(x, y), 3),
  cov_raw = round(cov(x, y), 3),
  cov_ownSD = round(cov(x/sd(x), y/sd(y)), 3),
  cov_commonSD = round(cov(x/sdX0, y/sdy0), 3))
knitr::kable(rbind(
  sc(E$deltaC_mean_mean, E$dag_mean, "29 edited"),
  sc(WC$deltaC_mean_mean, WC$dag_mean, "20 progenitors + Col-0"),
  sc(at16$Delta_13C, at16$FT16, "248 natural (FT16)"), row.names = FALSE)
```

set	r	cov_raw	cov_ownSD	cov_commonSD
29 edited	0.376	0.902	0.376	0.028
20 progenitors + Col-0	0.144	1.876	0.144	0.058
248 natural (FT16)	0.381	12.262	0.381	0.381

Do **not** mean-standardise $\delta^{13}\text{C}$ (Houle-style evolvability): its mean is negative on an arbitrary reference scale, so mean-scaled values are sign-inverted nonsense.

Relative variation (percentage)

$\delta^{13}\text{C}$ has a negative mean on an arbitrary (VPDB) zero, so a signed CV is negative and uninterpretable. We therefore report $100 * \text{SD} / |\text{mean}|$ for $\delta^{13}\text{C}$, and a conventional CV for flowering time, which has a true zero.

```
relsd <- function(x) 100 * sd(x) / abs(mean(x))
cvpct <- function(x) 100 * sd(x) / mean(x)
cvrow <- function(lab, x, y) data.frame(
  set = lab, n = length(x),
  d13C_relSD_pct = round(relsd(x), 2),
  FT_CV_pct = round(cvpct(y), 2))

cvtab <- rbind(
  cvrow("29 edited", E$deltaC_mean_mean, E$dag_mean),
  cvrow("20 progenitors + Col-0", WC$deltaC_mean_mean, WC$dag_mean),
  cvrow("19 progenitors", W$deltaC_mean_mean, W$dag_mean),
  cvrow("248 natural (FT16)", at16$Delta_13C, at16$FT16),
  cvrow("261 natural (FT10)", at10$Delta_13C, at10$FT10))
knitr::kable(cvtab, row.names = FALSE)
```

set	n	d13C_relSD_pct	FT_CV_pct
29 edited	29	1.44	24.49
20 progenitors + Col-0	20	1.40	43.59
19 progenitors	19	1.25	39.67
248 natural (FT16)	248	3.35	31.50
261 natural (FT10)	261	3.34	20.23

```
c(d13C_ratio_progenitors_over_edited = round(re1sd(WC$deltaC_mean_mean) /
                                             re1sd(E$deltaC_mean_mean), 2),
  FT_ratio_progenitors_over_edited   = round(cvpct(WC$dag_mean) /
                                             cvpct(E$dag_mean), 2))
```

```
## d13C_ratio_progenitors_over_edited  FT_ratio_progenitors_over_edited
##                                0.97                                1.78
```

Relative $\delta^{13}\text{C}$ variation is essentially identical between edited lines and progenitors (ratio ≈ 0.97), while relative flowering-time variation is $\sim 1.8\times$ lower in the edited lines — the same conclusion as the F tests, without the mean shift that inflates the raw 32-fold variance ratio.

Only the progenitors-vs-edited comparison is condition-controlled (same chamber, same experiment). The natural panels were phenotyped at 16 °C and 10 °C, and CV depends strongly on condition — the same accessions give 31.5% at FT16 and 20.2% at FT10 — so chamber-vs-atlas CV ratios are not interpretable.

Measured vs imputed

imputed_wt&mut_dc13.csv holds separate wild-type and flc-mutant predictions for 306 accessions. For the edited lines that were actually measured it contains the **measured** values (fidelity $r = 1$), so imputation changes nothing there.

```
d$acc <- firstnum(d$line); d$acc[d$label == "col"] <- 6909
m <- merge(d, im, by.x = "acc", by.y = "id", all.x = TRUE)
Ei <- subset(m, label == "mut"); Wi <- subset(m, label == "wt")
knitr::kable(rbind(
  stat(Ei$flc_mut_dc13,      Ei$dag_mean, "edited, imputed"),
  stat(Ei$deltaC_mean_mean, Ei$dag_mean, "edited, measured"),
  stat(Wi$wt_dc13,          Wi$dag_mean, "progenitors, imputed"),
  stat(Wi$deltaC_mean_mean, Wi$dag_mean, "progenitors, measured")), row.names = FALSE)
```

set	n	r	p	slope	slope_SE	cov	sd_d13C	sd_FT
edited, imputed	28	0.376	0.0489	4.58	2.22	0.935	0.452	5.51
edited, measured	29	0.376	0.0447	4.58	2.18	0.902	0.444	5.41
progenitors, imputed	19	0.138	0.5740	2.56	4.46	6.288	1.568	29.09
progenitors, measured	19	-0.066	0.7890	-5.07	18.66	-0.722	0.377	29.09

```
c(fidelity_edited = round(cor(Ei$flc_mut_dc13, Ei$deltaC_mean_mean, use = "complete"), 3),
  fidelity_wildtype = round(cor(Wi$wt_dc13, Wi$deltaC_mean_mean, use = "complete"), 3))
```

```
## fidelity_edited fidelity_wildtype
##            1.000            0.705
```

Scale warning. The two imputed columns are on different scales — mean wild-type -35.6% vs mean mutant -30.8% , an offset of $\sim 4.8\%$ — which is why the analysis code divides them by different constants (-37.21 and -31.14). This is a plausible contributor to the wild-type/mutant non-overlap Reviewer 1 questioned in Fig. 3C.

```
c(mean_wt_imputed = round(mean(im$wt_dc13), 2), sd_wt = round(sd(im$wt_dc13), 3),
  mean_mut_imputed = round(mean(im$flc_mut_dc13), 2), sd_mut = round(sd(im$flc_mut_dc13), 3),
  offset = round(mean(im$wt_dc13) - mean(im$flc_mut_dc13), 2))
```

```
## mean_wt_imputed      sd_wt mean_mut_imputed      sd_mut
##      -35.620         1.145       -30.840         0.283
##      offset
##      -4.780
```

Summary of the Fig. 3B/3C tests

Claim	Test	Result
Editing changed the correlation	Fisher z vs natural population	p = 0.97 — no
Editing changed the slope	interaction term	p = 0.68 — no
Editing changed the covariance	bootstrap	p = 0.84 — no
Editing collapsed flowering-time variance	F test	p = 6×10⁻⁴ — yes, 32-fold
Wild-type relationship in Fig. 3B is significant	correlation	p = 0.55 — no

The constraint is relaxed because flowering-time variance collapses, not because the genetic correlation between the traits changes.

Table S28 — evolvability, respondability, and multivariate constraint

The Hansen & Houle framework behind Table S28 — a walkthrough — plus the construction of the G matrix behind it.

```
# palette (colorblind-safe, validated)
col_beta <- "#2a78d6" # selection gradient (blue)
col_dz <- "#eb6834" # predicted response (orange)
col_aux <- "#1baf7a" # third series (aqua)
col_ell <- "#898781" # G ellipse / muted ink
col_ink <- "#0b0b0b"
```

The setup: one equation, four numbers

Everything in Table S28 is geometry applied to the multivariate breeder's equation (Lande 1979):

$$\Delta \bar{z} = G \beta$$

- β — the **selection gradient**: the direction (in 12-trait space) in which selection pushes the phenotype. From our multivariate selection analysis (one β per fitness measure: survival, fecundity, composite fitness).
- G — the **additive genetic variance–covariance matrix** of the same 12 traits.
- $\Delta \bar{z}$ — the **predicted response**: where the population mean actually moves in one generation.

The key point of Hansen & Houle (2008): selection *proposes* a direction, but G *disposes*. G acts on β like a lens — it stretches the response along directions rich in genetic variance and shrinks it along depleted ones, so $\Delta \bar{z}$ generally points somewhere *between* β and g_{max} (the leading eigenvector of G).

For a **unit-length** β (all formulas below assume $|\beta| = 1$), the framework defines:

Quantity	Formula	Meaning
Evolvability	$e(\beta) = \beta^\top G \beta$	additive genetic variance available in the direction selection favours
Respondability	$r(\beta) = G \beta $	how far the population moves, in any direction
Angle	$\theta = \angle(\beta, G \beta)$	how far the response is rotated away from where selection points
Conditional evolvability	$c(\beta) = (\beta^\top G^{-1} \beta)^{-1}$	variance along β when all orthogonal directions are held constant by stabilising selection

Quantity	Formula	Meaning
Autonomy	$a(\beta) = c(\beta)/e(\beta)$	fraction of the variance along β that is free of the rest of the phenotype

Table S28 reports the first three (plus the cosine of the angle). Conditional evolvability and autonomy are the rest of the framework, worth knowing.

The identity that ties them together

$$\cos \theta = \frac{\beta \cdot G\beta}{|\beta| |G\beta|} = \frac{\beta^\top G\beta}{|G\beta|} = \frac{e(\beta)}{r(\beta)}$$

So the cosine alignment is **not an independent third line of evidence** — it is exactly evolvability over respondability. Equivalently, the projection of the response onto the selection direction has length $r \cos \theta = e$. That gives the cleanest one-sentence intuition:

Respondability is how much the population moves; evolvability is how much of that movement is in the direction selection actually pushes. Everything else is dragged sideways by genetic covariances.

Check it against Table S28:

```
S28 <- data.frame(
  fitness      = c("Survival", "Composite fitness", "Fecundity (seeds)"),
  evolvability  = c(0.29752, 0.17818, 0.32536),
  respondability = c(0.50996, 0.28275, 0.79680),
  angle_deg    = c(54.30815, 50.93882, 65.90009),
  cos_align    = c(0.58343, 0.63015, 0.40833)
)
S28$e_over_r   <- with(S28, evolvability / respondability)
S28$angle_from_cos <- acos(S28$cos_align) * 180 / pi
knitr::kable(S28, digits = 3,
  caption = "Table S28 values, plus the internal identity:  $\cos \theta = e/r$ ,  $\theta = \text{acos}(\cos \theta)$ ."
```

Table 8: Table S28 values, plus the internal identity: $\cos \theta = e/r$, $\theta = \text{acos}(\cos \theta)$.

fitness	evolvability	respondability	angle_deg	cos_align	e_over_r	angle_from_cos
Survival	0.298	0.510	54.308	0.583	0.583	54.308
Composite fitness	0.178	0.283	50.939	0.630	0.630	50.939
Fecundity (seeds)	0.325	0.797	65.900	0.408	0.408	65.900

The last two columns reproduce the reported cosine and angle exactly — three numbers per row, two degrees of freedom.

Geometric intuition in two traits

A 2-trait toy makes the whole framework visible. The grey ellipse is the 1-SD contour of G ; blue is β (unit length); orange is $\Delta \bar{z} = G\beta$.

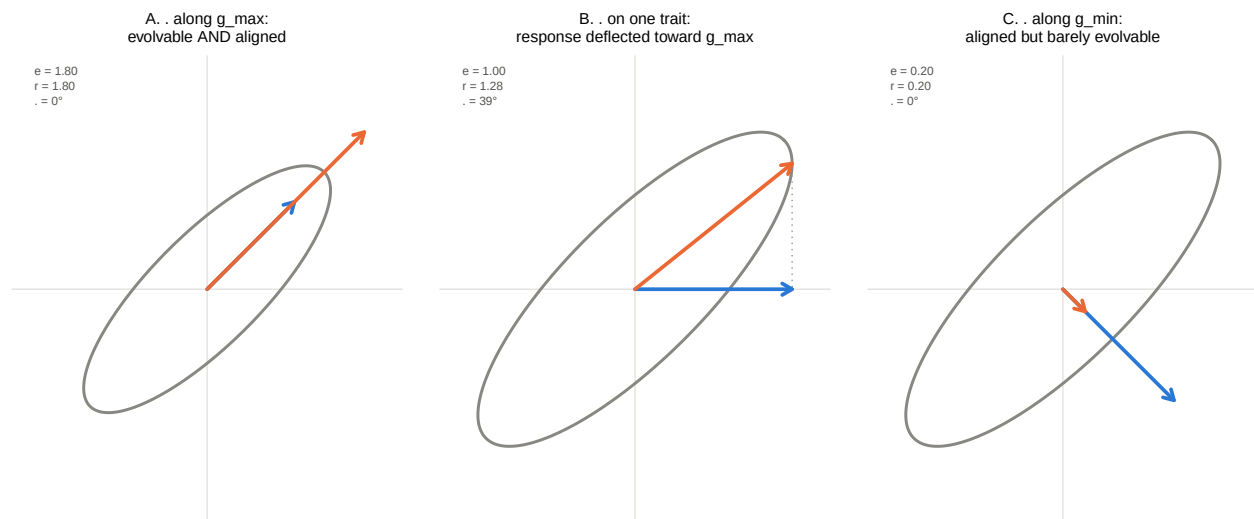
```
draw_case <- function(G, beta, main) {
  beta <- beta / sqrt(sum(beta^2))
  dz  <- as.vector(G %*% beta)
  e   <- as.numeric(t(beta) %*% G %*% beta)
```

```

r <- sqrt(sum(dz^2))
th <- acos(sum(beta * dz) / r) * 180 / pi
# 1-SD ellipse of N(0, G)
E <- eigen(G)
tt <- seq(0, 2 * pi, length.out = 200)
ell <- E$eigenvectors %*% diag(sqrt(pmax(E$values, 0))) %*% rbind(cos(tt), sin(tt))
lim <- max(abs(ell[,1]), abs(ell[,2])) * 1.15
plot(NA, xlim = c(-lim, lim), ylim = c(-lim, lim), asp = 1,
     axes = FALSE, xlab = "", ylab = "", main = main, font.main = 1, cex.main = 0.95)
abline(h = 0, v = 0, col = "#e5e4e0")
lines(ell[1, ], ell[2, ], col = col_ell, lwd = 2)
# projection of dz onto beta (dashed): its length IS e(beta)
proj <- sum(dz * beta) * beta
segments(dz[1], dz[2], proj[1], proj[2], lty = 3, col = col_ell)
arrows(0, 0, beta[1], beta[2], length = 0.08, lwd = 2.5, col = col_beta)
arrows(0, 0, dz[1], dz[2], length = 0.08, lwd = 2.5, col = col_dz)
legend("topleft", bty = "n", cex = 0.8, text.col = "#52514e",
      legend = c(sprintf("e = %.2f", e), sprintf("r = %.2f", r),
                  sprintf("θ = %.0f°", th)))
}

Gc <- rbind(c(1, 0.8), c(0.8, 1)) # two traits, r_g = 0.8
op <- par(mfrow = c(1, 3), mar = c(1, 1, 3, 1))
draw_case(Gc, c(1, 1), "A. β along g_max:\nevolvab AND aligned")
draw_case(Gc, c(1, 0), "B. β on one trait:\nresponse deflected toward g_max")
draw_case(Gc, c(1, -1), "C. β along g_min:\naligned but barely evolvable")

```



```
par(op)
```

The three panels are the three qualitatively different fates of a selection gradient (blue = β , orange = $\Delta \bar{z}$, dotted = projection of the response back onto β , whose length is e):

- **A — evolvable and aligned.** β points along g_{max} . The response is large and exactly where selection wants it: $e = r$, $\theta = 0$.
- **B — the deflection case (this is our survival/fitness situation).** Selection acts on one trait only, but the genetic correlation drags the correlated trait along: the response rotates toward g_{max} . Note $e < r$: the population moves further than the variance along β alone would allow — but partly in a direction nobody asked for. **With $r_g = 0.8$ between just two traits, the deflection is already 39° .** Our fecundity case (69°) is this phenomenon

compounded across 12 dimensions.

- **C — aligned but not evolvable.** β points along g_{min} — antagonistic to the correlation (e.g., selection to *increase* one trait while *decreasing* a positively correlated one). Because g_{min} is an eigenvector, there is no deflection at all ($\theta = 0$) — but e has collapsed to 0.2. **The angle alone can therefore look reassuring while evolution is nearly stalled.** You need e and θ together, which is why S28 reports both.

So *when* is something evolvable, and when is it not?

This is the conceptual payoff of the framework. “Evolvable” is not a property of a trait — it is a property of a **direction** in trait space.

1. Univariate heritability is almost never the binding constraint. Every one of our 12 traits has substantial genetic variance on its own (the diagonal of G — see below). Judged trait-by-trait, everything is evolvable. That criterion is too weak.

2. What matters is variance along β , and it is bounded by the eigenvalues: $\lambda_{min} \leq e(\beta) \leq \lambda_{max}$. If selection happens to point along a direction where G is thin — a *nearly-null subspace* — evolvability can approach zero even though every single trait is heritable. Directions of low variance arise precisely when selection is **antagonistic to the correlation structure** (panel C): wanting more of one thing and less of a genetically coupled other. That is the geometry of evolutionary conflict.

3. The expected evolvability of a random direction is the mean eigenvalue, $\bar{e} = \text{tr}(G)/n$. This gives a natural yardstick: is the direction selection actually favours richer or poorer in variance than a random direction would be? (We compute this for our G below — the answer is *poorer*.)

4. Conditional evolvability can be much smaller than evolvability. $e(\beta)$ assumes the rest of the phenotype is free to drag along. If the other directions are under stabilising selection (usually true), the usable variance is $c(\beta) \leq e(\beta)$, sometimes dramatically so when the focal direction is strongly integrated with the rest. A trait can look evolvable in isolation and be nearly frozen in context.

5. “Heritability is not evolvability” (Hansen, Pélabon & Houle 2011). $h^2 = V_A/V_P$ measures accuracy of parent–offspring resemblance, not capacity to change: a trait can have high h^2 and trivial usable variance, and vice versa, because V_P is in the denominator. Hansen & Houle therefore work with **mean-standardised** variance ($I_A = V_A/\bar{z}^2$, “percent change per unit selection”). Keep this in mind when reading our numbers — see the standardisation caveat at the end.

In one line: a phenotype is evolvable in the direction of selection when $e(\beta)$ is a healthy fraction of $\bar{e}$, and unconstrained when additionally θ is small and $c(\beta) \approx e(\beta)$. It is constrained when selection points into a variance-poor, strongly integrated direction — regardless of how heritable every individual trait is.

The real thing: our 12-trait G matrix

G combines two sets of genomic estimates from the 1001 Genomes panel: the **diagonal** holds each trait’s univariate GEMMA LMM PVE (Table S16) — its standardised additive genetic variance — and the **off-diagonals** come from bivariate GEMMA mvLMMs run for all 66 trait pairs, each contributing the scale-free genetic correlation $r_g = V_{g,12}/\sqrt{V_{g,11}V_{g,22}}$. The univariate runs provide the variances because they have the larger per-trait sample size; the mvLMMs contribute only the correlation structure. Every covariance is then $G_{ij} = r_{g,ij}\sqrt{PVE_i PVE_j}$:

$$G = D_{PVE}^{1/2} \text{nearPD}(R_g) D_{PVE}^{1/2}$$

The nearPD projection is needed because 66 independently estimated pairwise correlations need not be jointly positive semi-definite — and here they are not (raw R_g eigenvalues run to -0.31), even though every pair is individually legal; the projection moves no correlation by more than 0.11. This is the construction implemented in `regenerated-tables/regenerate_S21_S28_S29.R`; the chunk below builds it live, checking each log’s $V_g[2, 1]$ against the deposited matrix and the result against the generated file.

```

G_dep <- readRDS("data/Gmatrix_imp_multivar_corr_matrix.rda") # RDS despite .rda
traits <- rownames(G_dep) # full trait names, deposited order
short <- c("FT16", "Dormancy", "RootGR", "RootAngle", "Delta13C", "StomDens",
          "StomIdx", "VernGrowth", "Germ10C", "GrowthRate", "RGR", "ABA")
logs <- list.files("multivarGWAS/all_impGWAS_logoutput", pattern = "log.txt$",
                  full.names = TRUE)
stopifnot(length(logs) == 66)
Rg <- matrix(NA_real_, 12, 12, dimnames = list(traits, traits)); diag(Rg) <- 1
for (f in logs) {
  base <- sub("^imp_mGWAS_", "", sub("\\.log\\.txt$", "", basename(f)))
  t1 <- traits[startsWith(base, paste0(traits, "_"))][1]
  t2 <- substring(base, nchar(t1) + 2)
  ll <- readLines(f, warn = FALSE)
  i <- grep("REMLE estimate for Vg", ll)[1]
  v11 <- as.numeric(strsplit(trimws(ll[i + 1]), "\\s+")[1][1])
  v2 <- as.numeric(strsplit(trimws(ll[i + 2]), "\\s+")[1][1])
  stopifnot(abs(G_dep[t1, t2] - v2[1]) < 1e-6) # provenance: log == deposited
  Rg[t1, t2] <- Rg[t2, t1] <- v2[1] / sqrt(v11 * v2[2])
}
Rg_pd <- as.matrix(Matrix::nearPD(Rg, corr = TRUE)$mat)
Dp <- diag(sqrt(diag(G_dep))) # deposited diagonal = univariate PVE
G_final <- Dp %*% Rg_pd %*% Dp
dimnames(G_final) <- list(short, short)

# identical to the supplement's G
stopifnot(max(abs(G_final - readRDS("regenerated-tables/G_regenerated.rda"))) < 1e-12)
ev_final <- eigen(G_final, symmetric = TRUE)
round(ev_final$values, 4)

```

```

## [1] 3.3021 1.8799 0.7595 0.4159 0.3320 0.2118 0.2002 0.1853 0.1141 0.0180
## [11] 0.0000 0.0000

```

Constraint geometry with this G and the Table S13 selection gradients (Madrid low-water environment, mli):

```

s13 <- readRDS("data/multivariate_SelectionResults_targettraits_1122122.rda") # S13 betas
map <- c(ABA = "ABA", Delta13C = "Delta13C", Dormancy = "Dormancy",
        FloweringTime = "FT16", GerminationPerc = "Germ10C",
        GrowthRate = "GrowthRate", RGR = "RGR", RootRGR = "RootGR",
        RootHorizIndex = "RootAngle", StomataDensity = "StomDens",
        StomataSize = "StomIdx", Vernalization = "VernGrowth")
get_beta <- function(fit, env = "mli") {
  sub <- s13[s13$env == env & s13$fitness == fit, ]
  b <- setNames(sub$mean.x, map[sub$trait])[short] # mean.x = the beta column
  b / sqrt(sum(b^2))
}
geom <- function(Gm, b) {
  dz <- as.vector(Gm %*% b)
  e <- sum(b * dz); r <- sqrt(sum(dz^2))
  c(e = e, r = r, angle = acos(e / r) * 180 / pi)
}
fits <- c(Survival = "rSurvival_fruit", Fitness = "rFitness", Fecundity = "rSeeds")
betas <- lapply(fits, get_beta)
knitr::kable(round(t(sapply(betas, function(b) geom(G_final, b))), 3), caption =
  "G as built above, S13 mli gradients – matches regenerated-tables/S28_regenerated_mli.csv.")

```

Table 9: G as built above, S13 mli gradients — matches regenerated-tables/S28_regenerated_mli.csv.

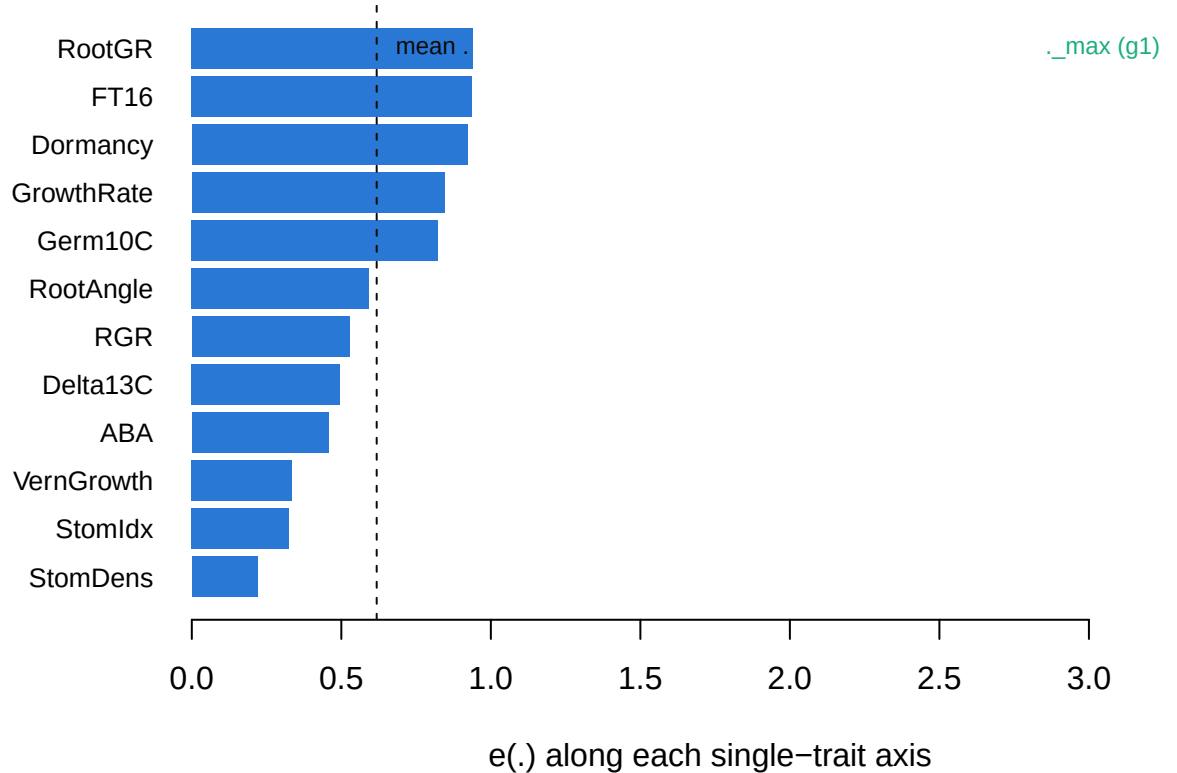
	e	r	angle
Survival	0.298	0.510	54.308
Fitness	0.178	0.283	50.939
Fecundity	0.325	0.797	65.900

Survival 54.3°, composite fitness 50.9°, fecundity 65.9° — these reproduce Table S28 exactly. One technical note for what follows: nearPD lands *on* the PSD boundary, so the two smallest eigenvalues of G are exactly zero (rank 10 of 12) — harmless for e , r , and θ , which never use G^{-1} , but decisive for the conditional measures below.

Evolvability along every single-trait axis vs. along the eigenvectors

Setting $\beta = (0, \dots, 1, \dots, 0)$ makes $e(\beta) = G_{ii}$: the diagonal of a variance-standardised G is each trait's own standardised additive variance — the heritability-scale values. Compare those to the eigenvalue range of G :

```
e_trait <- diag(G_final)
op <- par(mar = c(4, 7, 1, 1))
bp <- barplot(sort(e_trait), horiz = TRUE, las = 1, col = col_beta, border = NA,
              xlab = "e(β) along each single-trait axis", cex.names = 0.8,
              xlim = c(0, max(ev_final$values) * 1.05))
abline(v = mean(ev_final$values), col = col_ink, lty = 2)
abline(v = max(ev_final$values), col = col_aux, lty = 2)
text(mean(ev_final$values), max(bp), "mean λ", pos = 4, cex = 0.75, col = col_ink)
text(max(ev_final$values), max(bp), "λ_max (g1)", pos = 2, cex = 0.75, col = col_aux)
```



`par(op)`

Five of the twelve trait axes clear the mean eigenvalue (the $h^2 \approx 0.82$ – 0.94 block: root growth, flowering time, dormancy, growth rate, germination); the stomatal traits sit near 0.2. But even the lowest trait axis carries an order of magnitude more variance than the bottom eigendirections of G ($\lambda_{10} = 0.018$, $\lambda_{11} = \lambda_{12} \approx 0$) — judged univariately, every trait is evolvable (point 1 above). The interesting question is what happens along the directions selection actually favours.

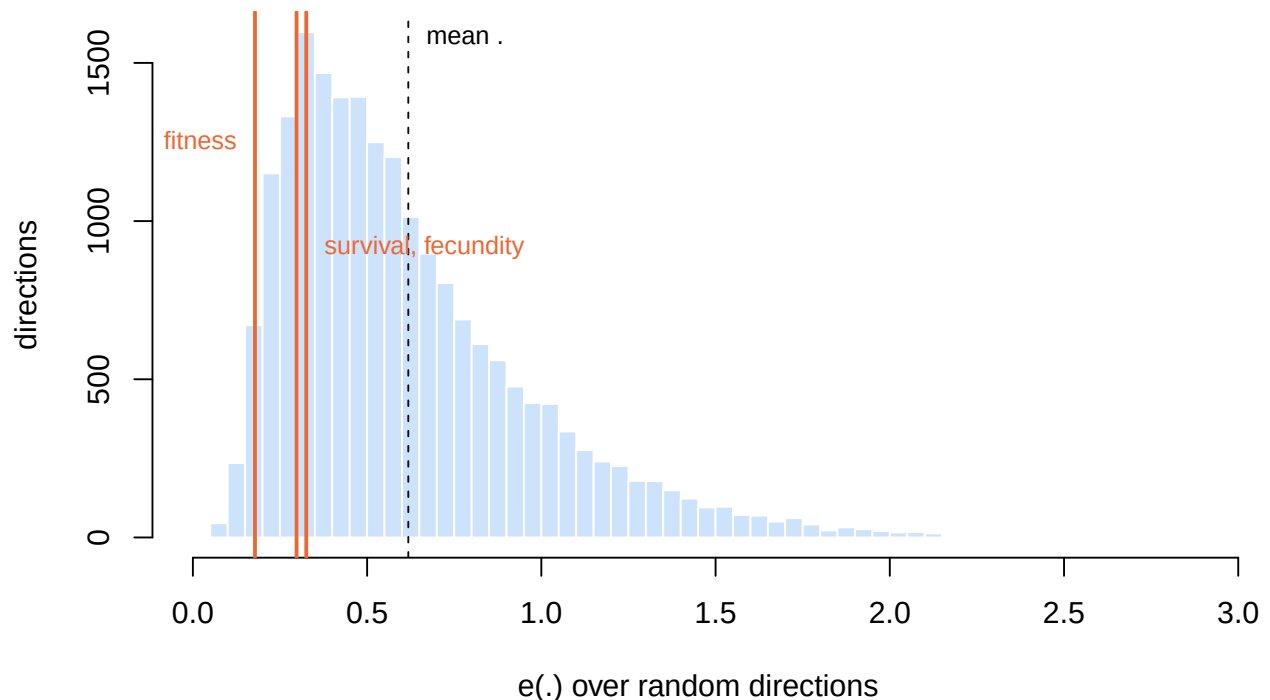
Where do the selection directions sit?

We sample thousands of random unit directions in the 12-dimensional trait space, compute $e(\beta)$ for each, and place the three selection directions in that distribution.

```
set.seed(1)
n <- 20000
X <- matrix(rnorm(n * 12), n, 12)
B <- X / sqrt(rowSums(X^2))           # random unit vectors
e_rand <- rowSums((B %*% G_final) * B) # beta' G beta, row-wise

obs <- sapply(betas, function(b) geom(G_final, b)["e"])
names(obs) <- names(fits)

op <- par(mar = c(4, 4, 1, 1))
hist(e_rand, breaks = 80, col = "#cde2fb", border = "white",
     main = "", xlab = "e(β) over random directions", ylab = "directions")
abline(v = mean(ev_final$values), col = col_ink, lty = 2)
abline(v = obs, col = col_dz, lwd = 2)
text(mean(ev_final$values), par("usr")[4] * 0.95, "mean λ", pos = 4, cex = 0.8)
text(min(obs), par("usr")[4] * 0.75, "fitness", pos = 2, cex = 0.8, col = col_dz)
text(max(obs), par("usr")[4] * 0.55, "survival, fecundity", pos = 4, cex = 0.8, col = col_dz)
```



```
par(op)

round(c(mean_lambda = mean(ev_final$values),
       sapply(obs, function(v) mean(e_rand < v))), 3)
```

```
## mean_lambda    Survival    Fitness    Fecundity
##           0.618         0.168         0.030         0.214
```

All three selection directions fall **below the mean** of the evolvability distribution — the composite-fitness gradient down in the bottom 3% of random bearings, survival and fecundity in the bottom sixth to fifth. Selection favours directions that are variance-poor relative to a random direction through this trait space. That is multivariate constraint stated as a percentile rather than an angle — a milder version of the same message the deflection angles carry. (Note the two measures disagree about *which* direction is worst off: fecundity has the largest deflection angle, but composite fitness sits in the thinnest part of the evolvability distribution. An angle measures where the response *goes*; a percentile measures how much variance the direction *has*.)

Conditional evolvability and autonomy (the part S28 doesn't show)

$c(\beta)$ needs G^{-1} — and here the construction of G matters. nearPD projected the raw R_g onto the boundary of the PSD cone, so G has two eigenvalues that are exactly zero: it is **singular**, rank 10 in 12 dimensions. The projected matrix asserts two trait combinations with literally zero genetic variance; every single-trait axis overlaps that null space, so $c(\beta) \approx 0$ for every trait. Bend the zero eigenvalues up with a floor before inverting, and the answer tracks the floor, not the data:

```
floors <- c(1e-4, 1e-2, 1e-1)
aut <- sapply(floors, function(fl) {
  v <- pmax(ev_final$values, fl) # eigenvalue floor
  Gi <- ev_final$vectors %*% diag(1 / v) %*% t(ev_final$vectors)
  (1 / diag(Gi)) / diag(G_final) # autonomy per trait axis
})
colnames(aut) <- paste0("floor_", floors)
round(aut, 3)
```

```
##           floor_1e-04 floor_0.01 floor_0.1
## FT16           0.009       0.176       0.295
## Dormancy       0.015       0.334       0.472
## RootGR         0.001       0.045       0.202
## RootAngle      0.002       0.045       0.261
## Delta13C       0.001       0.118       0.502
## StomDens       0.212       0.597       0.901
## StomIdx        0.001       0.078       0.426
## VernGrowth     0.000       0.037       0.353
## Germ10C        0.001       0.050       0.258
## GrowthRate     0.003       0.127       0.246
## RGR            0.002       0.099       0.390
## ABA            0.001       0.109       0.471
```

Median autonomy 0.001, 0.10, or 0.37 depending on an arbitrary regularisation constant: **conditional evolvability and autonomy are not estimable from this G** . A joint, rather than pairwise, re-estimation of G would be needed before any conditional claim can be made (e , r , and θ are untouched by this — they never use G^{-1}).

Reading the constraint results with all of this in hand

Table S28 — e , r , θ for the three fitness measures (S13 mli gradients):

Fitness measure	e	r	θ
Survival	0.298	0.510	54.3°
Composite fitness	0.178	0.283	50.9°
Fecundity	0.325	0.797	65.9°

- **Survival / composite fitness** are panel B of the toy figure: usable variance along β , response deflected 51–54°, so ~58–63% of the movement is where selection points ($\cos \theta = 0.58, 0.63$) — real but moderate constraint.
- **Fecundity** is panel B sliding toward panel C: 65.9°, by far the worst-aligned. Only ~41% of the response is along the selection direction, yet fecundity has the *largest* response vector ($r = 0.80$) — the population moves the most while accomplishing the least. A big r with a modest e is not power, it is leakage: G takes the fecundity gradient and spends much of it on correlated side-effects.
- The constraint is strongest for exactly the fitness component that opposes survival in our selection analyses — the geometry of G is entangled with the survival–fecundity conflict the paper is about: the fecundity gradient puts ~64% of its squared weight on the four variance-poorest eigendirections of G , versus ~37% for survival (panel C logic).
- And the punchline for intuition: the median pairwise $|r_g|$ is 0.36, with a quarter of pairs above 0.55 — no single pair, on its own, explains a 51–66° deflection. **Constraint is a property of the whole matrix, not of any pair** — a dozen moderate-to-strong correlations compound. Judging constraint from a correlation table understates it.

Caveats specific to our numbers

1. **Standardisation.** Hansen & Houle’s canonical e is *mean-standardised* — they divide by the trait mean ($I_A = V_A/\bar{z}^2$), so e reads as % change per generation per unit selection. Our traits are z-scored (mean-centred and SD-scaled); the centring is immaterial here, since variances and covariances are shift-invariant, so effectively our G is *variance-standardised* (SD scale) — chosen because $\delta^{13}\text{C}$ has a negative mean on an arbitrary reference scale, making mean-scaling meaningless. Consequence: our e values are comparable *within* this analysis (across the three fitness measures, against the eigenvalue yardstick) but not directly against the published distribution of mean-scaled evolvabilities in the Hansen & Houle compilations.
2. **Pairwise assembly.** G is assembled from 66 separately estimated bivariate models rather than one joint 12-trait model (computationally prohibitive at this scale). The nearest-PSD projection this requires moves individual correlations by up to 0.11 and places G on the PSD boundary (rank 10 of 12), which is why conditional measures are not reported.
3. **No uncertainty.** All geometric quantities here are point estimates — eigenvalues, angles, and especially conditional measures deserve confidence intervals (e.g., by resampling the pairwise r_g and h^2 estimates), which do not exist yet.
4. **Constancy of G .** $\Delta\bar{z} = G\beta$ is a one-generation prediction. Over many generations selection itself erodes and rotates G , so “constraint” here means *bias in the immediate response*, not a permanent wall.

References. Lande 1979 (doi:10.1111/j.1558-5646.1979.tb04678.x); Hansen & Houle 2008, *J. Evol. Biol.* (doi:10.1111/j.1420-9101.2008.01573.x); Hansen, Pélabon & Houle 2011, “Heritability is not evolvability” (doi:10.1007/s11692-011-9127-6); Agrawal & Stinchcombe 2009 on constraint from G (doi:10.1098/rspb.2008.1855).